

10.6.4.3. Data

Upon [path expansion](#) of the value in **Path**, the hierarchy and structure of the encoded data for each concrete Path SHALL be based on the schema description of the specified attribute within the specified cluster. The TLV encoding of each element in the data SHALL follow the rules of encoding as provided in [Data Types](#).

10.6.4.3.1. Lists

The various values in the **Change** enumeration are realized as follows:

Change Type	Realization
REPLACE	Path SHALL refer to a list with ListIndex omitted and Data SHALL contain new values that will replace the existing contents of the list.
ADD	Path SHALL refer to a list with ListIndex containing a value of null and Data containing the new value of the list item that will be added to the list.

NOTE

ListIndex is currently only allowed to be omitted or null. Any other value SHALL be interpreted as an error.

The below bullets detail the suggested patterns for mutating a list, depending on the encoding of the list.

- A single [AttributeDataIB](#) containing a path to the list itself and Data that contains all items in the list encoded as a TLV array. This option SHOULD be selected if it is possible to encode the entirety of the list in a single [AttributeDataIB](#) that fits in a single message.

This option SHOULD be selected when sending AttributeReports as a part of a ReportDataMessage, when it is possible to encode the entirety of the list in a single [AttributeDataIB](#) that fits in a single message.

This option SHOULD also be selected when modifying the ACL or Extension attribute of the [Access Control Cluster](#) as part of a WriteRequestMessage, when it is possible to encode the entirety of the list in a single [AttributeDataIB](#) that fits in a single message.

This option SHALL NOT be used in a WriteRequestMessage for other list attributes.

- A series of [AttributeDataIBs](#), with the first containing a path to the list itself and Data that is an empty array, which signals clearing the list, and subsequent [AttributeDataIBs](#) each containing a path to each list item with **ListIndex** being null, in order, and Data that contains the value of the list item.
- A series of [AttributeDataIBs](#), with the first containing a path to the list itself and Data that is an array that contains some number of list items, which signals replacing the list with those items, and subsequent [AttributeDataIBs](#) each containing a path to each list item with **ListIndex** being null, in order, and **Data** that contains the value of the list item.